Routing & Wireless Concepts Project 2026

Student Name: Kamil Salih

Student number: C00307549

Group: Cyber Group B

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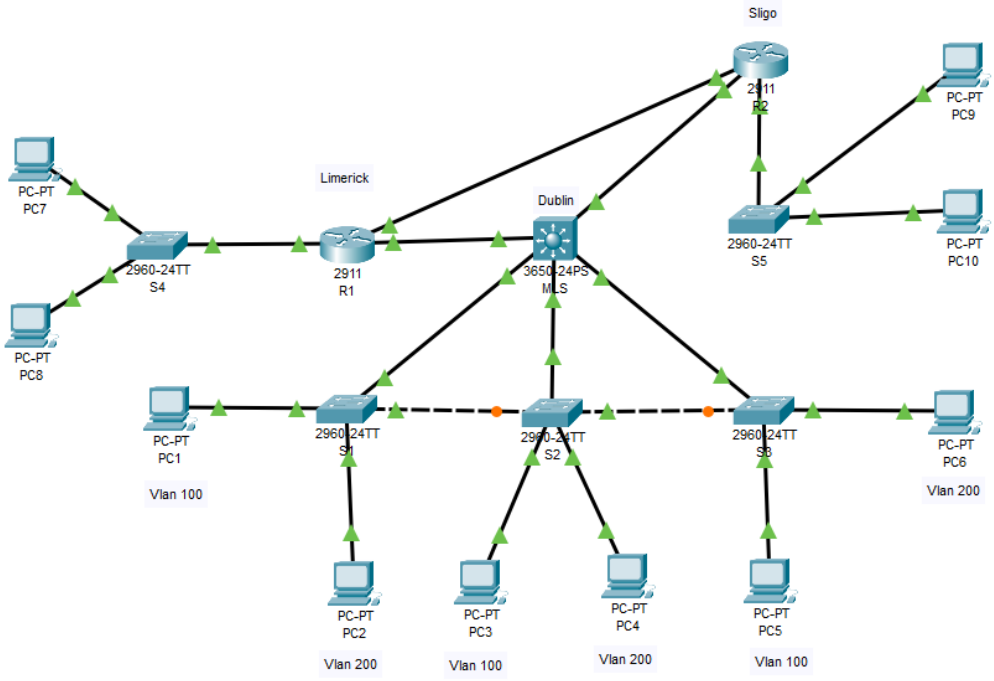
Subject: Networking: Wireless and Routing Concepts

Introduction

Aurora Healthcare Group operates a central campus in Dublin and two regional clinics in Limerick and Sligo. The aim of this project is to design a secure, scalable, and efficient LAN-based internetwork that enables communication between all sites. The design focuses on VLAN segmentation, secure routing, and protection against common LAN vulnerabilities.

Network topology overview

Labelled network diagram



Explanation of the overall topology

The network consists of a Central Campus (Dublin) connected to two remote clinics (Limerick and Sligo) via WAN links.

At the Central Campus, a Layer 3 device (MLS) performs:

1. Inter-VLAN routing
2. DHCP services
3. Static and floating static routing

Multiple Layer 2 switches (S1, S2, S3, S4, S5) connect end devices.

Redundant links exist between switches to support STP.

At the remote clinics, each site contains:

1. One router (R1 for Limerick, R2 for Sligo)
2. One switch
3. End devices

Each router acts as a DHCP server for its local network.

The WAN link connections:

- Dublin to/from Limerick
- Dublin to/from Sligo
- Redundant paths allow failover using floating static routes

Requirements mapping

Small Table

Requirement	Devices/configurations used
1. VLAN Design at Central Campus	VLAN 100 (Clinical) and VLAN 200 (Admin) were configured on all Central Campus switches with trunk links between switches to enable inter-VLAN routing.
2. IP Addressing and DHCP	DHCP was configured on the MLS (Central Campus) and on the routers at each clinic using the appropriate subnet ranges.
3. Layer 2 Security at Central Campus	Port security, DHCP snooping, Dynamic ARP Inspection, BPDU Guard implemented.
4. Secure Remote Administration	SSH configured on MLS with local user accounts and Telnet has been disabled.

5. Static and Floating Static Routing Between Sites	Static routes were configured on the MLS, R1, and R2 to enable communication between all networks. Floating static routes with higher administrative distance were configured as backup paths to ensure connectivity in the event of a primary link failure.

LAN and Layer 2 security

Two VLANs were implemented at the Central Campus:

1. VLAN 100 (Clinical): 10.10.100.0/25
2. VLAN 200 (Admin): 10.10.200.0/27

These VLANs segment traffic to:

1. Improve performance
2. Enhance security
3. Separate sensitive clinical data from administrative systems

Inter-VLAN routing is performed on the Multi-Layer Switch (MLS) using SVIs (Switched Virtual Interfaces), allowing communication between VLANs when required.

To protect the Central Campus network from LAN-based attacks, the following Layer 2 security mechanisms were implemented:

Port Security

Limits the number of MAC addresses per port. Prevents MAC flooding attacks by restricting unauthorized devices

DHCP Snooping

Identifies trusted and untrusted ports. Blocks rogue DHCP servers and prevents DHCP starvation attacks

Dynamic ARP Inspection (DAI)

Validates ARP packets against the DHCP snooping database. Prevents ARP spoofing and man-in-the-middle attacks

BPDU Guard

Applied to access ports. Prevents rogue switches from affecting STP topology

VLAN Security Measures

Proper trunk configuration (only VLANs 100 and 200 allowed). Unused ports disabled. Reduces risk of VLAN hopping attacks

Together, these features significantly improve the security posture of the LAN by mitigating several attack vectors. These configurations directly mitigate MAC flooding, DHCP attacks, ARP spoofing, VLAN hopping, and STP manipulation as outlined in the project requirements.

Scalability, security and reliability

Central Campus:

- VLAN 100: 10.10.100.0/25
- VLAN 200: 10.10.200.0/27
- DHCP configured on MLS to dynamically assign IP addresses

Limerick Clinic:

- Network: 172.16.10.0/27
- DHCP configured on Router R1

Sligo Clinic:

- Network: 172.16.20.0/27
- DHCP configured on Router R2

Dynamic addressing ensures:

1. Efficient IP management
2. Reduced manual configuration errors
3. Scalability for future growth

The network has been designed with future growth, security, and resilience in mind.

Scalability

VLAN-based design allows easy addition of new departments or users, without major reconfiguration. DHCP simplifies device deployment without manual configuration. Additional clinics can be integrated by assigning new subnets and routes

Security

VLAN segmentation isolates sensitive clinical and administrative traffic. Layer 2 security features protect against internal network attacks. SSH ensures secure remote administration

Reliability

Redundant links between switches enable STP to prevent loops while maintaining backup paths. Floating static routes provide automatic failover if a primary WAN link fails. Ensures continuous communication between sites even during network issues (WAN link fails)